

Supplementary Material for “Semantic burstiness across principal directions of textual variation”

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S1. Inter-event-time distributions for the remaining representative texts

Inter-event-time distributions for the three representative texts other than *War and Peace*, whose corresponding figure appears in the main text. Equivalent figures for every text of the corpus, at any threshold quantile, can be regenerated with `scripts/run_text.py` in the code repository.

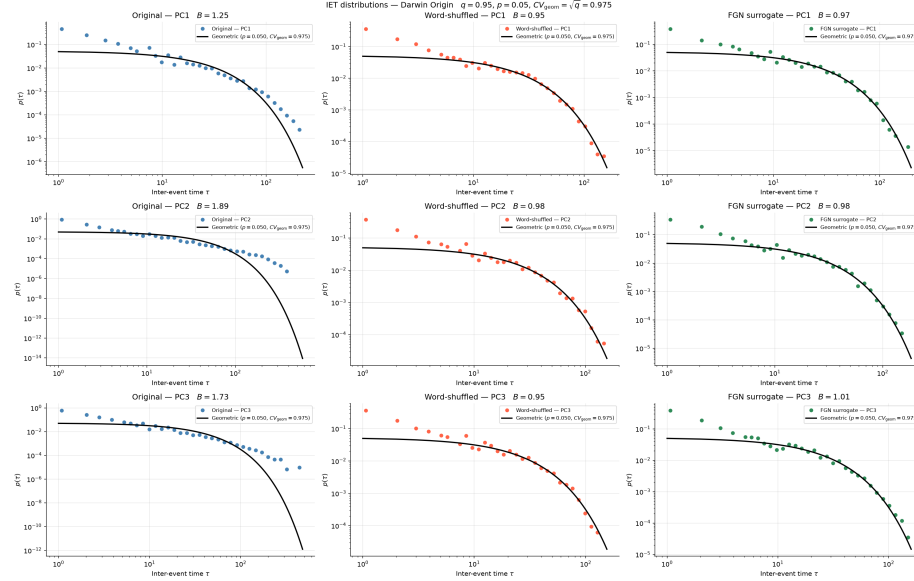


Figure S1: Inter-event time distributions for PC1, PC2, PC3 of *On the Origin of Species* ($q = 0.95$). Same conventions as the corresponding figure in the main text.

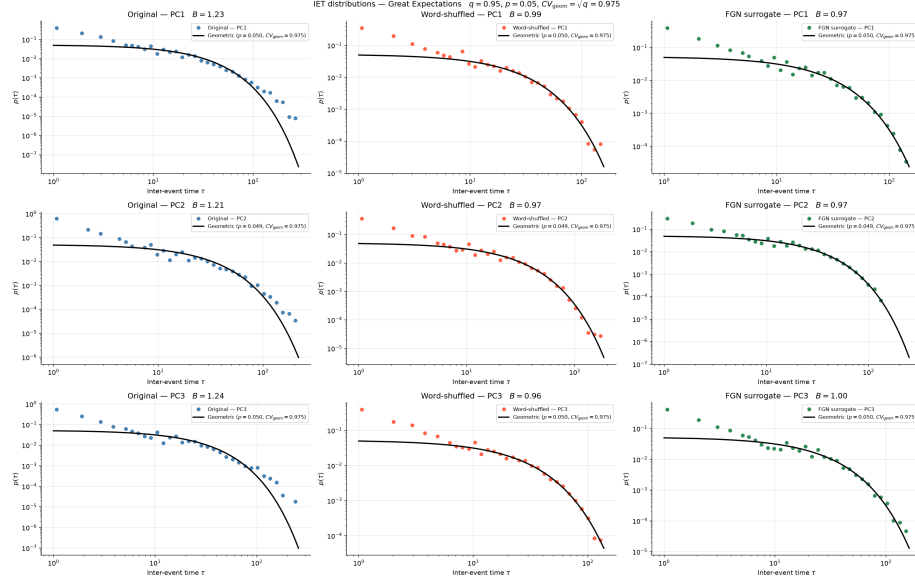


Figure S2: Inter-event time distributions for PC1, PC2, PC3 of *Great Expectations* ($q = 0.95$). Same conventions as the corresponding figure in the main text.

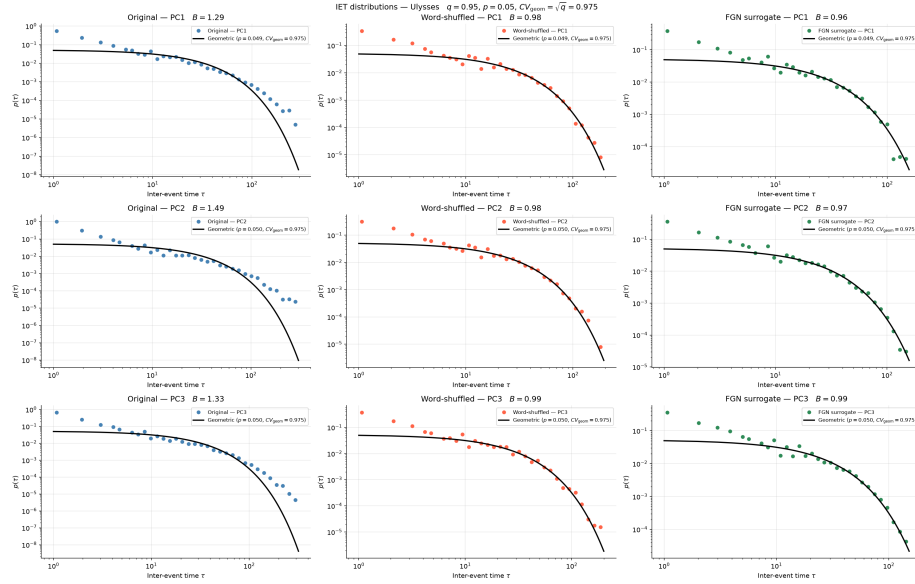


Figure S3: Inter-event time distributions for PC1, PC2, PC3 of *Ulysses* ($q = 0.95$). Same conventions as the corresponding figure in the main text.

S2. Additional robustness figures

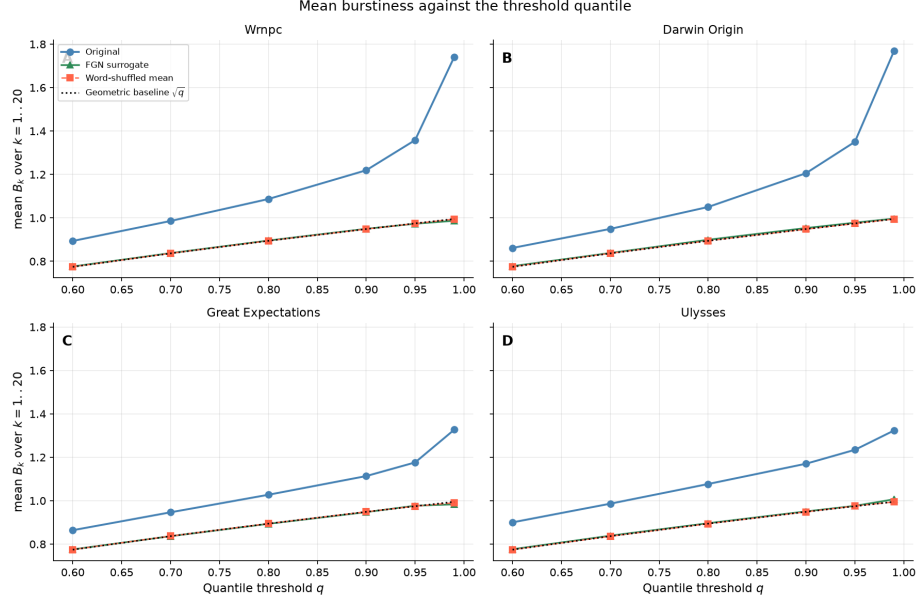


Figure S4: Mean burstiness over $k = 1, \dots, 20$ as a function of the threshold quantile q , for the four representative texts: **A** *War and Peace*, **B** *On the Origin of Species*, **C** *Great Expectations*, **D** *Ulysses*. The original stays above both controls and above the geometric baseline $\sqrt{q}$ throughout $q \in [0.60, 0.99]$, and the full ordering $B_k^{\text{orig}} > B_k^{\text{FGN}}$ and $B_k^{\text{orig}} > B_k^{\text{surr}}$ holds for all 20 components at every displayed q in each text. Absolute burstiness grows towards more extreme thresholds, as rare visits to the upper tail accentuate episode-to-episode clustering.

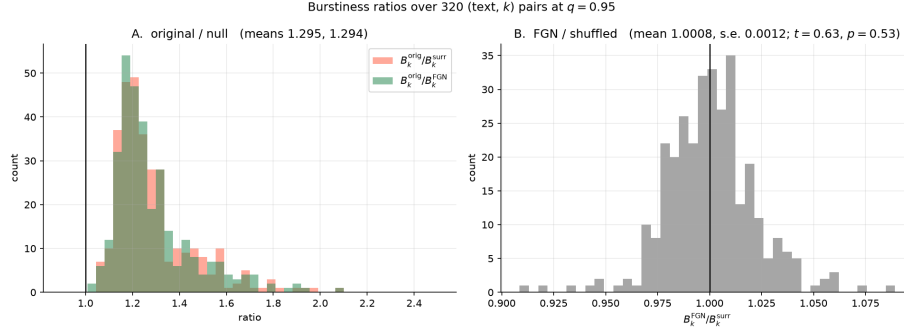


Figure S5: **Distribution of burstiness ratios over 320 (text, k) pairs at $q = 0.95$.** **Panel A:** distributions of $B_k^{\text{orig}}/B_k^{\text{surr}}$ (red) and $B_k^{\text{orig}}/B_k^{\text{FGN}}$ (green), both concentrated well above 1. The two distributions are nearly identical, confirming that the two null models are equivalently far below the original. **Panel B:** distribution of $B_k^{\text{FGN}}/B_k^{\text{surr}}$, tightly centred on 1 (mean 1.0008, s.e. 0.0012; $t = 0.63$, $P = 0.53$), statistically indistinguishable from unity, confirming $B_k^{\text{FGN}} \approx B_k^{\text{surr}}$ across the entire corpus.

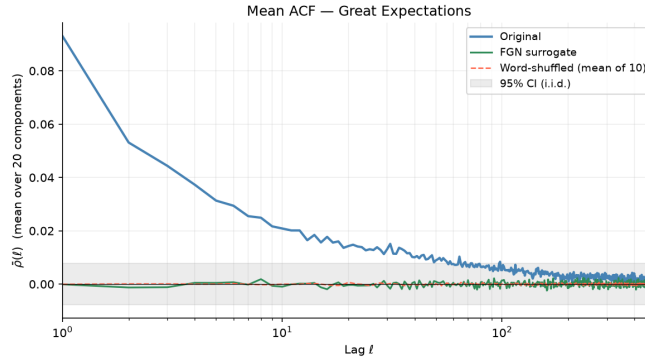


Figure S6: **Mean ACF $\bar{\rho}(\ell)$ averaged over all $K = 20$ PCA components — *Great Expectations* (Dickens).** Same colour conventions as the per-component ACF figure in the main text. The original text maintains positive mean autocorrelation well beyond the i.i.d. band across the full range of lags shown. The FGN surrogate and word-shuffled null are indistinguishable from the i.i.d. reference at all lags, confirming that rank-level long-range memory (present in the FGN by construction) does not propagate to the semantic projection space.

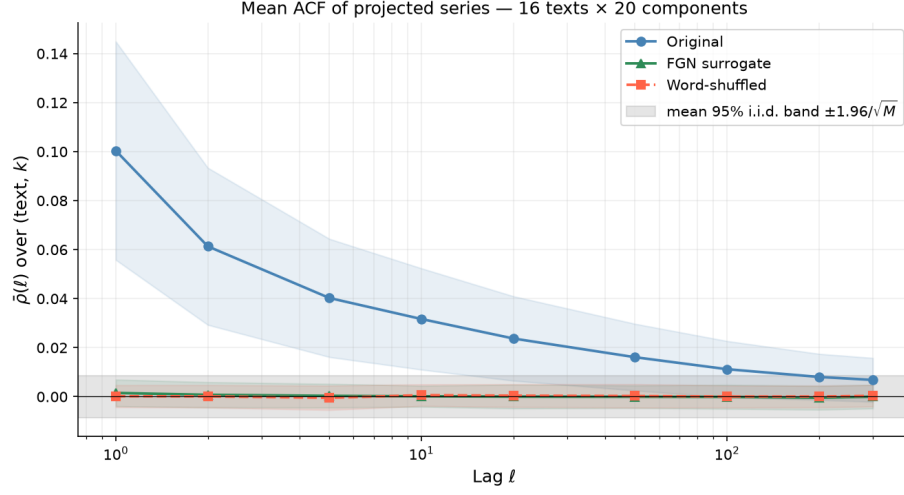


Figure S7: **Mean ACF $\bar{\rho}(\ell)$ of projected series across the corpus of 16 texts and $K = 20$ components, as a function of lag ℓ (log scale).** Shaded bands: ± 1 standard deviation over (text, k) pairs. Grey band: mean 95% i.i.d. confidence band $\pm 1.96/\sqrt{M}$. Blue circles: original filtered text. Green triangles: FGN surrogate. Red squares: word-shuffled null. The original maintains significant positive ACF across all lags tested ($\ell = 1, \dots, 300$). Both null models are indistinguishable from zero at all lags.

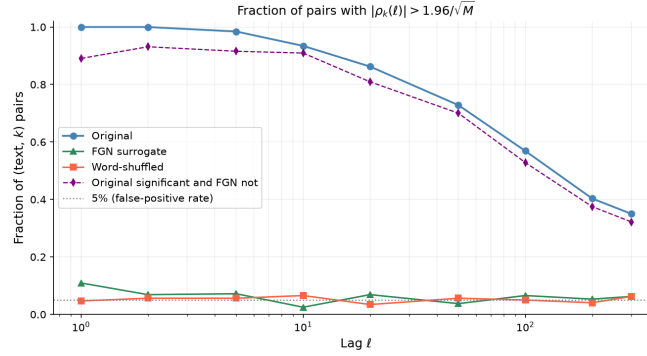


Figure S8: **Fraction of text–component pairs with significant ACF as a function of lag ℓ (log scale).** Blue: original; green: FGN; red: word-shuffled; purple dashed: original significant and FGN not. The original remains clearly separated from both controls through lags of order 10^2 . The short-lag FGN excess is concentrated disproportionately in PC1, the component most affected by the embedding-geometry artefact discussed in the preprocessing section of the main text.

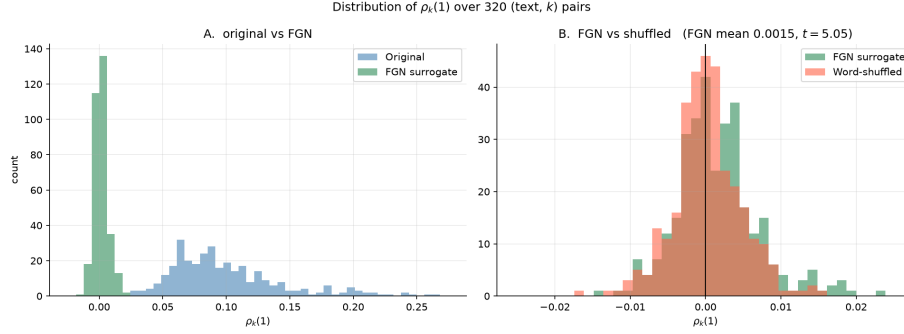


Figure S9: **Distribution of $\rho_k(1)$ over 320 (text, k) pairs.** **Panel A:** original (blue) and FGN (green). The two distributions are completely separated: every original pair exceeds its own text-specific i.i.d. band (which ranges from 0.004 to 0.015 across the corpus), while nearly all FGN pairs fall inside it. **Panel B:** FGN (green) and word-shuffled (red) on an expanded scale. Both are centred near zero; a t -test gives $t = 5.05$ for $\bar{\rho}^{\text{FGN}}(1) = 0$ but the effect size is negligible (0.002) relative to the orig-null gap (0.098).

S3. Full numerical tables

Table S1: ACF of projections at lags $\ell = 1$ and $\ell = 100$ for *Great Expectations*. The i.i.d. reference band is $\pm 1.96/\sqrt{M} = \pm 0.0077$. Values clearly outside the band are in bold; values that round to the band edge are left unmarked.

k	$\rho_k(1)$		$\rho_k(100)$	
	Orig	FGN	Orig	FGN
1	0.060	0.004	0.008	0.004
2	0.208	0.007	0.015	0.004
3	0.136	-0.002	0.018	0.004
4	0.123	-0.003	0.008	-0.003
5	0.086	-0.002	0.004	-0.004
6	0.121	0.004	0.005	-0.001
7	0.127	0.002	0.007	-0.003
8	0.103	0.001	0.004	0.003
9	0.082	-0.002	0.008	0.001
10	0.083	0.004	0.003	0.003
$(k=11..20 \text{ all have } \rho^{\text{FGN}}(1) \leq 0.007)$				

Table S2: DFA scaling exponents α_k of projected series $x_t^{(k)}$ for *Great Expectations* (Dickens, $\alpha = 0.689$). Every original value exceeds the shuffled reference by more than 2σ (the spread over the ten shuffled realisations is $\sigma \simeq 0.004$), and every FGN value lies within 0.05 of 0.5.

k	α^{orig}	α^{FGN}	α^{surr}	k	α^{orig}	α^{FGN}	α^{surr}
1	0.605	0.511~	0.501	11	0.57	0.506~	0.500
2	0.681	0.500~	0.491	12	0.642	0.493~	0.502
3	0.693	0.518~	0.495	13	0.564	0.512~	0.500
4	0.634	0.514~	0.499	14	0.595	0.484~	0.497
5	0.64	0.486~	0.496	15	0.645	0.486~	0.497
6	0.633	0.515~	0.501	16	0.579	0.505~	0.494
7	0.647	0.504~	0.498	17	0.586	0.511~	0.493
8	0.633	0.469~	0.497	18	0.617	0.515~	0.507
9	0.643	0.482~	0.494	19	0.595	0.489~	0.505
10	0.630	0.482~	0.503	20	0.622	0.495~	0.502
Mean				$\bar{\alpha}^{\text{orig}} = 0.623 \pm 0.034$ $\bar{\alpha}^{\text{FGN}} = 0.499 \pm 0.014$ $\bar{\alpha}^{\text{surr}} = 0.499 \pm 0.004$			